

# THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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## ABSTRACT

We study how physical scale, population size, and sensing shape the emergence of complex behavior in open-ended ecological environments. Our setting deliberately removes the usual reward-design loop: agents receive no explicit objective, learn no supervised target, and persist only through reproduction, mutation, and selection. Because agents also change the resources and population around them, each policy evolves inside a dynamic ecology rather than against a fixed task. The contribution of this paper is therefore not a single optimized controller, but an empirical account of which behaviors become visible when objective-free evolutionary dynamics are run at large environmental and population scale. Our first contribution is scale: using modern hardware and a new multi-agent simulator, we run worlds far larger than prior settings of this kind, reaching more than 60,000 agents, each with its own evolved neural network policy. Our second contribution is behavioral: across these worlds, we identify emergent long-range resource extraction, vision-based foraging, and predation under survival and competition pressures. Our third contribution is an analysis of when these behaviors appear: sensing modalities and environmental scale strongly affect both emergence and stability, with some behaviors appearing only in sufficiently large environments and populations and becoming more consistent as scale increases. Together, these results suggest that large ecological simulations are a promising instrument for studying machine learning systems in an era of increasingly abundant computation and efficient accelerator-based frameworks. Experimental code will be released upon acceptance.

## 1 INTRODUCTION

The emergence of biodiversity and complex behavior through evolution has long been studied in evolutionary theory (Darwin 1859; Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). Across these fields, models of emergence range from speciation and adaptive radiation (Coyne & Orr 2004; Schluter 2000) to mathematical population genetics (Fisher 1930; Wright 1931). This paper takes inspiration from that tradition but asks a reviewer-facing machine learning question: what behaviors become measurable when evolving neural policies are placed in ecological worlds large enough for population-level effects to matter?

Natural ecosystems provide the motivation but also expose the experimental barrier. Mathematical models, laboratory studies, and field work have clarified many mechanisms of evolution, yet large natural ecosystems are difficult to control, instrument, and repeat. Controlled interventions at ecosystem scale can also damage or displace the very populations being studied. Our contribution is to use simulation not as a replacement for biology, but as a controlled testbed for isolating how scale, sensing, and ecological feedback affect behavioral emergence.

We therefore study the open-ended evolution of neural network policies in large-scale ecological simulations. Agents have no specified reward signal or learning objective; instead, they collect resources in order to survive and reproduce, and policy variation enters only through mutation when parents pass policies to children. This setup foregrounds the paper’s central contribution: it lets us evaluate how competitive and environmental pressures alone can produce behaviors, without optimizing agents toward a hand-designed task.

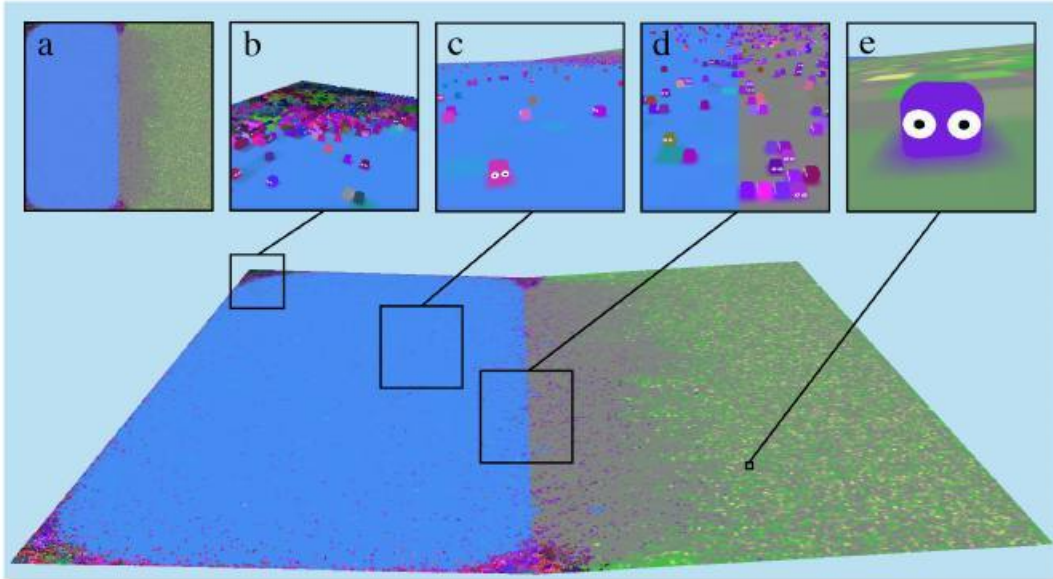


Figure 1: Overview of the ecological simulation used in this paper. The  $512 \times 512$  map visualization shows the full terrain in inset (a), a clustered group of agents in inset (b), agents in water in inset (c), agents moving between land and water in inset (d), and a close-up of resource gathering in inset (e).

To make these dynamics visible, we use a new JAX-based simulation environment that can evaluate large grid worlds rapidly on modern hardware. Our largest experiments contain more than 60,000 individual agents and more than 1,000,000 grid cells. Agents navigate to find food and other resources, and agents that gather enough resources reproduce by creating mutated copies of themselves. Figure 1 shows a 3D visualization of this environment. The methodological contribution is the combination of objective-free evolutionary dynamics, individually parameterized neural policies, and physical worlds large enough to support long-range ecological feedback.

Our goal is not only to better understand ecological phenomena through simulation, but also to study ecology as a mechanism for producing machine intelligence. Recent work in AI has shown that scale can transform model capabilities rather than merely improve them (Wei et al. 2022a); related work on chain-of-thought prompting suggests that some reasoning strategies become visible only beyond scale thresholds (Wei et al. 2022b). Yet many embodied-learning settings still involve relatively small environments and small populations. We ask the analogous ecosystem-level question: when we scale not one model but an entire evolving population, which strategies appear only once environmental size and complexity are sufficiently rich? This framing connects our empirical results to the broader claim that ecological scale may be a source of qualitatively different artificial behaviors.

Our experiments make three contributions explicit. First, in terrains with directional structure, agents with a simple compass sensor learn long-range resource-gathering expeditions that agents without the compass rarely sustain, and this behavior becomes more reliable at larger scales. Second, in ocean environments, agents with vision evolve more effective foraging and more precise attacking behavior than blind agents. Third, across both settings, larger environments and populations make the observed behaviors more stable and consistent. These findings show that the scale and complexity enabled by modern accelerators can reveal ecological behaviors that are difficult to observe in smaller simulations.

## 2 RELATED WORK

Artificial-life research provides the closest conceptual foundation for our contribution. Conway’s Game of Life (Games 1970) showed that simple local update rules can generate population-level structure, while Tierra (Ray 1992) and Avida (Brown) modeled evolving self-replicating programs. Similarly, Reynolds (1987) demonstrated that local motion rules can produce flocking and schooling.



Our work follows this lineage of studying emergence from simple rules, but shifts the emphasis to large populations of neural agents in resource-based environments where scale and sensing can be varied experimentally.

Evolutionary computation has also been used to design virtual agents and their components. Prior systems evolve locomotion morphology (Sims, 1994a,b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013), sensing (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025), and neural architectures (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). These works establish that evolution can search over agent bodies, sensors, and policies. Our paper contributes a complementary study: rather than using an objective or reward to produce better adapted agents, we examine which behaviors arise from ecological pressure itself and how those behaviors depend on environmental scale and complexity.

A second related thread uses evolution as an optimization engine. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) adapt populations of candidate solutions, and novelty-oriented variants encourage behavioral diversity (Lehman & Stanley, 2011; Cully et al., 2015). Similar approaches have optimized neural networks for Atari and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). Our experiments use mutation and selection, but the contribution is not a new optimizer and not improved performance on a fixed benchmark. Instead, we treat evolutionary mechanisms as the substrate for studying objective-free emergence in interacting populations.

A third thread studies organisms, communication, and autocurricula inside simulated environments. Early open-ended systems include Bedau (1992), Yaeger et al. (1994), Channon & Dampier (1998), and Tisue et al. (2004). More recent work grows organisms with emergent morphologies and behaviors using cellular automata (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025), studies emergent language and communication through Mordatch & Abbeel (2018) and Park et al. (2023), and explores autocurricula in which agents create learning pressure for one another through (Leibo et al., 2019) and smaller-population environments (Leibo et al., 2018; Team et al., 2021). Other systems evolve environments rather than agents, including Wang et al. (2019) and Aki et al. (2024), showing that environmental complexity can elicit sophisticated behavior. Our contribution is to keep the environment ecological and resource-driven while directly measuring how agent sensing and world scale change the behaviors that emerge.

The most closely related systems simulate populations of agents interacting in complex worlds. Baker et al. (2019) study hide and seeker agents trained with reinforcement learning to use tools, while other work trains populations for video games (Jaderberg et al., 2019), object manipulation (Petrenko et al., 2021), or capture tasks (Zheng et al., 2017; Yamada et al., 2020). These systems demonstrate important forms of emergence, but they usually retain fixed tasks and optimized objectives. Neural MMO (Suarez et al., 2019, 2023) is more open-ended, optimizing survival in a massively multi-agent game, but its population is capped at a fixed number of concurrent agents and does not include generational inheritance. Hamon et al. (2023) is closer to our setting because it includes variable population size, genetically inspired evolution, and generational timelines, though its environment is smaller, less feature-rich, and not focused on scale. Lu et al. (2024) also studies resource-gathering agents, but at a smaller maximum population of 256 agents and with an emphasis on programmable robot tool use. Relative to these systems, our contribution is the integration of naturally fluctuating populations, objective-free evolution, rich resource dynamics, and environment sizes large enough to expose collective ecological patterns.

Finally, our implementation builds on recent work emphasizing simulator speed and scale through optimized game-like environments (Koyamada et al., 2023; Matthews et al., 2024; Lechner et al., 2023). These systems show that modern accelerator-oriented software can change the feasible scale of multi-agent experimentation. We use similar JAX-based acceleration to make the empirical contribution of this paper possible: running ecological worlds with tens of thousands of independently parameterized neural agents and measuring the resulting emergent behaviors across scales.



### 3 METHOD

#### 3.1 ECOLOGICAL GAMES

We frame the simulator as an **Ecological Game (EG)** so that the methodological contribution is explicit: the formalism keeps Markovian multi-agent dynamics while removing fixed rewards and adding population change. As in Markov decision processes (MDPs) and stochastic games (SGs) (see Sutton et al. (1998) and Albrecht et al. (2024)), an EG includes a set of agents  $I$ , a state space  $S$ , an initial state distribution  $\rho_0$ , an action space  $A$ , and a transition function  $T$  mapping states  $s_t$  and actions  $a_{t,i}$  to a distribution over next states  $s_{t+1}$ . Unlike an SG, an EG removes the reward function and adds a population function  $\Gamma(s_t)$  that identifies which agents in  $I$  are alive and records parent identities, allowing birth and death events to be inferred. As with MDPs and SGs, the formalism can be partially observable; we refer to this case as a POEG. A POEG restricts each agent from observing the full state and instead uses an observation function  $O(s_t, i)$  that maps a state and player to that player’s observation. Our experiments instantiate a POEG whose observation function changes with the sensor configurations described next.

#### 3.2 ENVIRONMENT

The experimental environment is the concrete POEG used to test the paper’s claims about scale, sensing, and emergent behavior. It is a large grid world containing resources that agents must collect to survive and reproduce. The state is factored as  $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$ , where  $s_t^m$  stores map-level 2D data such as terrain and resource distributions, and  $s_t^a$  stores agent-level data such as position, health, and internal resources.

**Terrain:** Terrain supplies the physical structure that makes long-range ecological behavior possible. Each map has a rock layer defining grid-cell height, and water is initially placed on cells whose rock value lies below a preset sea level. During an episode, water flows downhill according to a discrete flow simulation based on this rock layer. We use the six global terrain configurations in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. To measure scale effects directly, we run  $64 \times 64$ ,  $128 \times 128$ ,  $256 \times 256$ ,  $512 \times 512$ , and  $1024 \times 1024$  grids and scale the initial population in proportion to grid area. After initialization, population size is not fixed; it fluctuates according to ecological dynamics.

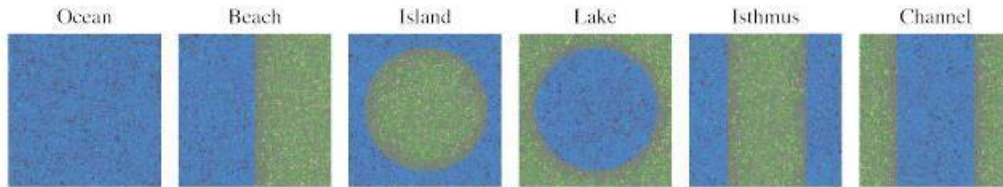


Figure 2: Terrain configurations used to test the paper’s scale and sensing claims, shown at  $256 \times 256$  grid size. Small dark spots are agents, and green and yellow regions indicate energy and biomass concentrations. Each snapshot is captured 5000 steps into an episode, after most water biomass has been eaten but before agents have evolved to survive on land for longer periods.

**Resources and Agent Health:** Resources define the survival pressures that replace explicit rewards. Agents must collect **water**, **energy**, and **biomass** to survive and reproduce. Biomass and energy are initially distributed randomly, but agents can consume and redistribute them through their movement, eating, death, and reproduction.

Each action expends a small amount of energy and water. Spent water returns to the landscape and flows downhill to the nearest body of water, whereas spent energy is destroyed. Biomass supports reproduction: every agent requires a fixed amount of biomass to exist, so reproduction succeeds only after an agent has collected enough excess biomass to donate to its offspring. Total biomass, like water, remains constant over the simulation; unlike water, it does not flow downhill or move without agent intervention. Energy is not conserved and slowly grows on free-standing biomass. These resource rules are a deliberate part of the contribution structure: they create a harder and more ecological problem than collecting a single food source. Appendix F.2 ablates the resource rules.



Agents maintain health points (HP), which connect resource management to survival and predation. If an agent overspends a critical resource, its health deteriorates rapidly, though it can spend energy and water to recover HP over time. Agents may also attack other agents, reducing the targets' HP. Age imposes a small but increasing damage cost, so agents eventually die of old age if no other cause kills them first. When an agent's HP reaches zero, its carried resources return to the environment at its grid cell. Thus, a successful attack can turn another agent's resources into food, creating a basic form of predation. Appendix F.8 ablates HP hyperparameters.

**Sensors:** Sensor variants are the main intervention used to test how perceptual information changes emergent behavior. Agents can receive four sensors: an **internal** sensor for age, health, and carried resources; an **external** sensor for resources in the current grid cell; a **compass**, inspired by magnetoreceptors in birds and other animals, that reports global direction as a 4-way one-hot vector; and **vision**, a  $7 \times 7$  top-down rendered patch around the agent with three color channels and a fourth channel for local elevation difference. Appendix F.6 ablates vision configurations.

**Actions:** The action space is intentionally small so that observed complexity can be attributed to ecological pressure rather than to a large hand-engineered controller. Agents choose from five action categories. **Rest** has no external effect and costs the least energy. The three **Move** actions rotate clockwise, rotate counterclockwise, or move forward one grid cell; if multiple agents attempt to occupy the same cell, the move is canceled. The **Attack** action kills all agents in the  $3 \times 3$  square directly in front of the attacker, when attacks are enabled; Appendix F.7 ablates attack strength. Three **Eat** actions correspond to the three resource types, and each transfers a fixed amount of the matching resource from the current cell into the agent's stomach up to a fixed capacity.

The final action is **Reproduce**, which creates an offspring through single-parent reproduction. Reproduction succeeds only if the parent has accumulated enough resources to donate to the child. The child is placed directly behind the parent, receives a mutated copy of the parent's policy weights, and immediately inherits a fixed amount of the parent's resources. This mechanism makes lineage and population turnover central to every experiment.

In the experiments, we disable selected sensors or the attack action to isolate the contribution of agent abilities. The three sensing combinations are internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**). We use **+A** to indicate that attacking is enabled; for example, (**RC+A**) denotes agents with resource sensors, a compass, and attack capability.

### 3.3 POLICIES

Each agent is controlled by its own neural network policy, making the population a collection of individually parameterized organisms rather than copies of a single shared controller. Policies are implemented as small, memoryless MLPs; removing memory keeps the computational model simple and reduces the per-agent parameter count, which is important for scaling to tens of thousands of agents. Raw sensor values are flattened, normalized to  $[-1, 1]$ , and passed through separate linear layers with output dimension 64. These representations are summed and processed by a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer outputs one logit per action, and actions are sampled from a softmax distribution with a temperature that evolves with the network weights. We store weights as half-precision bfloat16 values to support larger populations and faster runs. Depending on the sensors used, each agent policy contains roughly 10k-25k parameters. Appendix F.5 reports ablations on network size and architecture.

Initial policy weights use Kaiming initialization (He et al., 2015), and initial biases are zero. When an agent is born, its policy is copied from the parent and mutated by adding Gaussian noise with standard deviation  $3 \times 10^{-2}$  to all mutable weights. This mutation mechanism follows the spirit of prior neuroevolution work (Such et al., 2017) while intentionally avoiding more complex mechanisms such as crossover, so that the experiments foreground environmental scale and ecological pressure. Empirically, the low-precision setting required a larger mutation rate than we initially expected; Appendix F.4 evaluates this sensitivity.

Agents also carry a 3-channel color trait used when rendering them onto the map. This trait mutates in the same way as network weights and determines how an agent appears to other agents with

vision. In principle, color gives visual agents a channel through which different agent types could be distinguished, while preserving the same objective-free evolutionary framework.

### 3.4 COMPUTATION AND REPRODUCIBILITY

The computational contribution is that all experiments are implemented in JAX (Bradbury et al., 2018) and run on Nvidia H100 and A40 GPUs using a state-of-the-art simulation environment. Unless otherwise noted, every experiment runs for 2M environment steps or until extinction. Per-seed wall-clock time ranges from under an hour for  $64 \times 64$  grids to approximately 10 hours for  $1024 \times 1024$  grids on one H100 GPU. Appendix E gives additional simulator details, scaling measurements, and reproducibility considerations.

## 4 EXPERIMENTS

The experiments are organized around the paper’s main empirical contributions: whether scale and sensing change which objective-free ecological behaviors emerge, how consistently they appear, and whether the resulting population dynamics differ from smaller or less capable baselines.

### 4.1 LONG-DISTANCE RESOURCE GATHERING

The first contribution is evidence that a minimal navigation sensor can enable long-distance resource gathering, but that the behavior becomes reliable only at sufficient scale. In terrains with clear directional structure, agents with a compass (**RC**) develop substantially more effective long-range resource gathering than agents with only resource sensors (**R**). We refer to this behavior as mining because agents repeatedly travel from water onto land, collect biomass left by earlier deaths, and return to water.

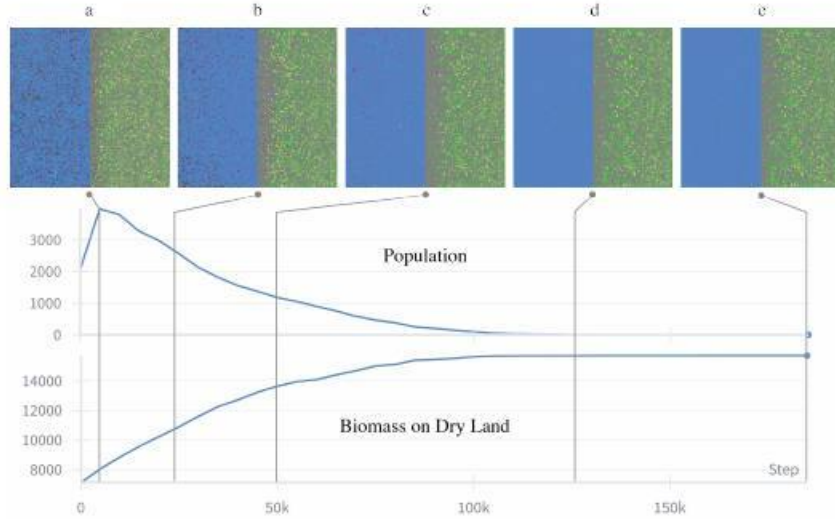


Figure 3: Representative failure case for long-distance resource gathering with (**R**) agents in a  $256 \times 256$  **Beach** environment. The top chart reports population over time, and the bottom chart reports free biomass on dry land. Without a compass, biomass accumulates on land while the population collapses.

Figure 3 illustrates why this task is ecological rather than merely navigational. After 5000 time steps (a), all initial dry-land agents have died from lack of water, while water agents have expanded rapidly because early resources are abundant. At this point, the population has not yet evolved policies that seek resources efficiently. From the early population peak, some water agents occasionally wander onto land (b), but they usually fail to return to the ocean and die, depositing biomass inland. By 50,000 steps (c), this process has nearly doubled the uneaten biomass on land, and the water population shrinks because the biomass needed to support it has been displaced. By 125,000 steps



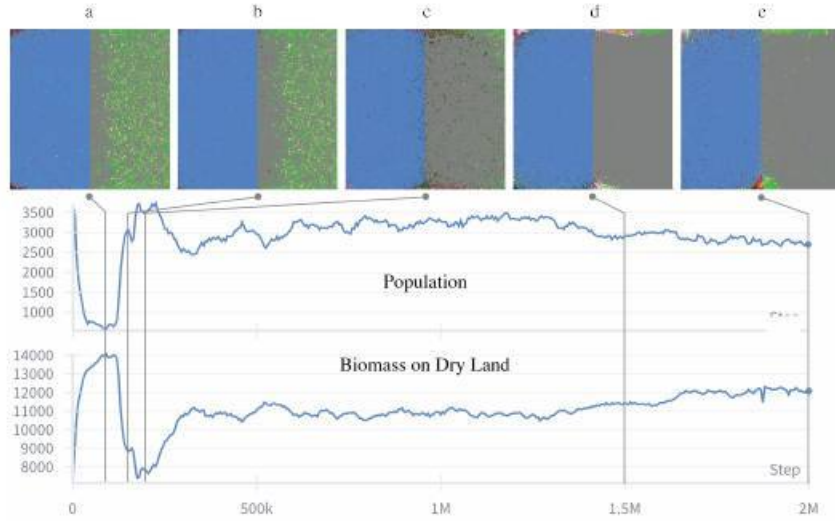


Figure 4: Representative emergence of long-distance mining with **(RC)** agents in a  $256 \times 256$  **Beach** environment. The top chart reports population over time, and the bottom chart reports free biomass on dry land. With a compass, agents eventually mine inland biomass and stabilize the population.

(d), almost all biomass has moved onto land and the population is close to extinction; the last agent dies around 185,000 steps (e).

This failure mode motivates the mining contribution. A local inability to return to water changes the global resource distribution, and those changes feed back into population survival. The same ecological catastrophe also creates a niche: any lineage that can travel onto land, exploit the accumulated biomass, and return safely to water gains access to a large resource reserve.

Figure 4 shows the corresponding contribution of the compass sensor. Early dynamics are similar: the population contracts and free biomass on land rises (a). However, some **(RC)** agents use the compass to drift toward the relative safety of corners. Around 150,000 steps (b), agents begin mining the land biomass by navigating between land and water. By 200,000 steps (c), these miners have cleared the beach. The population later forms clumps around edges (d), but it remains largely stable through the full 2,000,000-step run (e).

Table 1: Summary evidence for the long-distance mining contribution in the **Beach** world. Populations with a compass **(RC)** are compared to populations without a compass **(R)** across five world sizes and matched initial populations. reports the number of seeds with a mining event, and reports the number of seeds extinct before 2M steps. Table 3 in Appendix A gives seed-level details.

|             | 1024 × 1024 |     | 512 × 512 |     | 256 × 256 |     | 128 × 128 |     | 64 × 64 |     |
|-------------|-------------|-----|-----------|-----|-----------|-----|-----------|-----|---------|-----|
|             | 32768       |     | 8192      |     | 2048      |     | 512       |     | 128     |     |
| <b>(RC)</b> | 4/4         | 0/4 | 4/4       | 0/4 | 4/4       | 0/4 | 3/4       | 1/4 | 2/4     | 3/4 |
| <b>(R)</b>  | 1/4         | 0/4 | 0/4       | 4/4 | 0/4       | 4/4 | 0/4       | 4/4 | 0/4     | 4/4 |

We quantify this behavior across **Beach** worlds of size  $1024 \times 1024$ ,  $512 \times 512$ ,  $256 \times 256$ ,  $128 \times 128$ , and  $64 \times 64$ , with initial populations of 32768, 8192, 2048, 512, and 128 agents. Each condition is run with four random seeds. We count a mining event when free biomass on dry land drops by at least 10% from its historical maximum, and we also record the onset time of the event and the percentage biomass drop. Table 1 summarizes mining and extinction frequencies, and Table 3 in Appendix A reports seed-level details. The result is the first main empirical contribution: mining emerges consistently in **(RC)** agents and rarely in **(R)** agents. Scale also matters: mining occurs in

all (RC) seeds from  $256 \times 256$  upward, while smaller worlds show less stable populations and more early extinctions.

To test whether this finding is specific to the **Beach** terrain, we repeat the compass comparison on **Island**, **Lake**, **Isthmus**, and **Channel** maps at  $512 \times 512$  with an initial population of 8192. Table 2 gives the aggregate result, and Table 4 in Appendix A provides seed-level data. The same contribution pattern holds across terrain variants: compass agents generally mine more often and go extinct less often than agents without a compass, although the difficulty of mining depends on terrain geometry. Figure 12 in Appendix D provides a visual example of mining in the island terrain.

Table 2: Terrain-general evidence for the long-distance mining contribution. In **Island**, **Lake**, **Isthmus**, and **Channel** terrains at  $512 \times 512$ , compass agents (RC) are compared to agents without a compass (R).  reports seeds with a mining event, and  reports seeds extinct before 2M steps. Table 4 in Appendix A gives seed-level details.

|      | Island                                                                            |                                                                                   | Lake                                                                              |                                                                                   | Isthmus                                                                           |                                                                                   | Channel                                                                           |                                                                                     |
|------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|      |  |  |  |  |  |  |  |  |
| (RC) | 4/4                                                                               | 0/4                                                                               | 2/4                                                                               | 2/4                                                                               | 4/4                                                                               | 0/4                                                                               | 3/4                                                                               | 1/4                                                                                 |
| (R)  | 3/4                                                                               | 1/4                                                                               | 2/4                                                                               | 4/4                                                                               | 1/4                                                                               | 3/4                                                                               | 0/4                                                                               | 4/4                                                                                 |

We further analyze extinction and mining timing with additional seeds and survival models in Appendix B. Those results show that larger worlds and compass access both extend survival and accelerate mining emergence. Together, the experiments establish long-distance resource gathering as a scale- and sensor-dependent emergent behavior. They also show why the environment must be treated as an ecology: agents do not simply collect resources, but relocate them, create niches for later lineages, and determine whether the population survives long enough for those niches to be exploited.

## 4.2 VISUAL FORAGING AND PREDATION

The second empirical contribution is that vision changes both foraging and predation in the **Ocean** world. We compare blind agents with resource sensors, compass, and attacks (RC + A) to visual agents with the same abilities plus vision (RCV + A). Because both groups can attack and have the same explicit survival pressures, differences in population stability, biomass use, movement, eating, and attack precision isolate the effect of visual sensing.

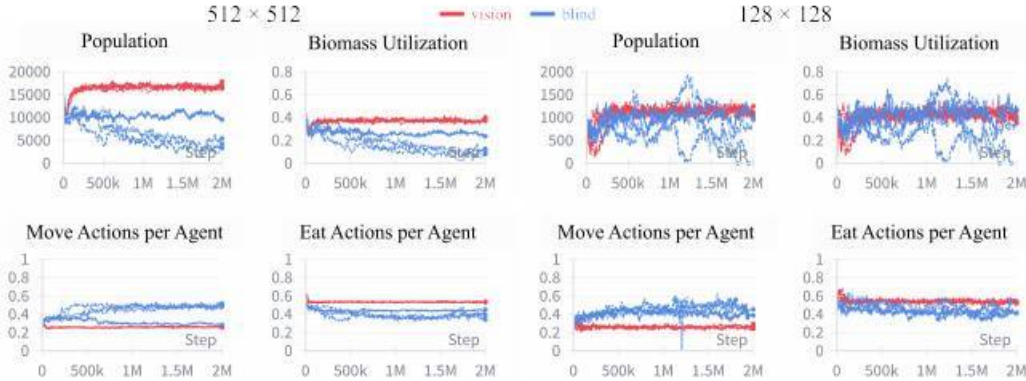


Figure 5: Foraging contribution of vision in  $512 \times 512$  and  $128 \times 128$  **Ocean** worlds. Each plot shows four vision-agent seeds (RCV + A) in red and four blind-agent seeds (RC + A) in blue. Population, biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Vision produces larger and more stable populations, greater biomass utilization, less movement, and more frequent effective eating, with the clearest separation at larger scale. Figure 9 in Appendix C gives the  $1024 \times 1024$  and  $256 \times 256$  comparison.



Figure 5 shows that visual agents maintain larger and more stable populations than blind agents. Vision also increases biomass utilization, increases eat-action frequency, and decreases movement, indicating that agents with local visual information forage more efficiently. The scale effect is again visible: the  $512 \times 512$  setting shows a clearer and more stable advantage for vision than the  $128 \times 128$  setting. Figure 9 in Appendix C extends the comparison to larger scales, and Table 5 in Appendix A reports the corresponding seed-level population and foraging metrics.

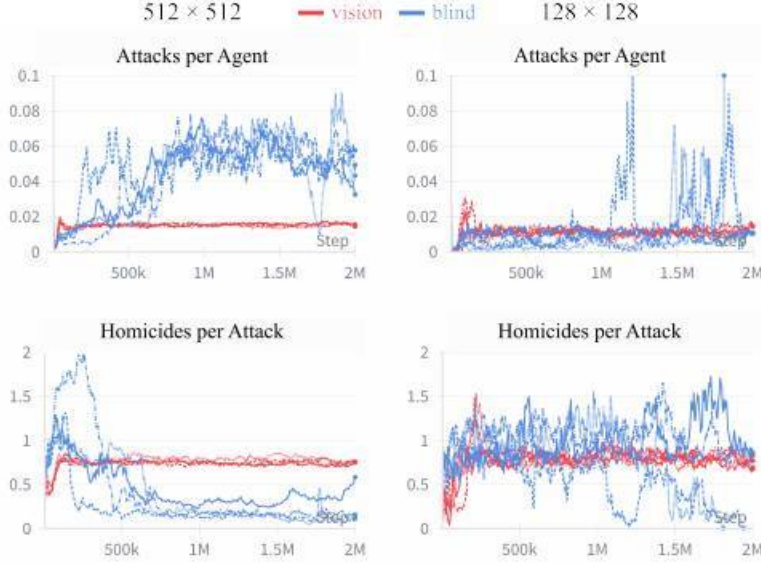


Figure 6: Predation contribution of vision in  $512 \times 512$  and  $128 \times 128$  **Ocean** worlds. Each plot shows four vision-agent seeds (**RCV + A**) in red and four blind-agent seeds (**RC + A**) in blue. Attacks per agent and homicides per attack are averaged across agents. Visual agents attack less often but kill much more precisely, suggesting specialized predation; the separation is strongest at larger scale. Figure 10 in Appendix C gives the  $1024 \times 1024$  and  $256 \times 256$  comparison.

Figure 6 shows the predation side of the same contribution. Visual agents attack less frequently but achieve far higher homicides per attack, consistent with more selective and precise attacking. In the larger **Ocean** worlds, blind agents often form dense clusters where frequent attacks can build up locally, a pattern that is rare for visual agents. This suggests that vision supports both finding resources and avoiding or targeting other agents; Figure 13 in Appendix D shows a representative visual comparison. In smaller environments the gap between visual and blind agents is less pronounced, but visual populations remain more stable.

### 4.3 ABLATIONS

Appendix F tests whether the main contributions are robust to simulator choices. We ablate initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8). We also run the analysis in Appendix F.3 to check that the large-world effects are not simply a self-averaging artifact. Together, these ablations position the main results as scale- and sensing-driven ecological effects rather than as consequences of a single hyperparameter choice.

## 5 CONCLUSION

This paper contributes evidence that open-ended ecological simulations can be a useful paradigm for studying the emergence of intelligence. Rather than optimizing agents for an explicit task, the experiments let neural policies evolve through survival, reproduction, mutation, and ecological feedback. Across worlds of varying scale and complexity, we find that environmental scale and sensory richness shape the emergence and stability of sophisticated behavior. Compass agents develop long-distance mining more reliably than agents without directional information, visual agents forage more

effectively and attack more precisely than blind agents, and larger environments make these effects more consistent. Taken together, the results support the paper’s central contribution: rich behaviors can evolve in large populations on a reasonable hardware budget without explicit objectives. They also motivate further study of ecology as both a scientific tool for understanding artificial behavior and a possible mechanism for developing increasingly capable artificial agents.

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## A ADDITIONAL DATA

This appendix collects the seed-level tables behind the main contribution statements. The summaries in the main text foreground the scale, compass, vision, foraging, and predation effects, while the tables here expose the per-seed measurements supporting those claims.

Table 3: Seed-level mining and extinction data supporting the **Beach** scale contribution. Compass agents (**RC**) are compared to agents without a compass (**R**) across five world sizes.  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum indicating the start of the mining event. For runs with multiple mining events, we report the first one.  denotes the timestep at which the run went extinct, or n/a if it lasted for 2M steps.

|                 | 1024x1024                                                                         |                                                                                    |                                                                                     |                                                                                     | 512x512                                                                           |                                                                                    |                                                                                     |                                                                                     | 256x256                                                                           |                                                                                    |                                                                                     |                                                                                     | 128x128                                                                           |                                                                                    |                                                                                     |                                                                                     | 64x64            |  |  |  |
|-----------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------|--|--|--|
|                 | 32768                                                                             |                                                                                    |                                                                                     |                                                                                     | 8192                                                                              |                                                                                    |                                                                                     |                                                                                     | 2048                                                                              |                                                                                    |                                                                                     |                                                                                     | 512                                                                               |                                                                                    |                                                                                     |                                                                                     | 128              |  |  |  |
| Compass On (RC) |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                  |  |  |  |
| Seed            |  |   |    |    |  |   |    |    |  |   |    |    |  |   |    |    |                  |  |  |  |
| 0               | ✓                                                                                 | 35k                                                                                | 11.1                                                                                | n/a                                                                                 | ✓                                                                                 | 105k                                                                               | 33.                                                                                 | n/a                                                                                 | ✓                                                                                 | 145k                                                                               | 49.3                                                                                | n/a                                                                                 | ✓                                                                                 | 105k                                                                               | 72.5                                                                                | n/a                                                                                 | x                |  |  |  |
| 1               | ✓                                                                                 | 90k                                                                                | 16.2                                                                                | n/a                                                                                 | ✓                                                                                 | 50k                                                                                | 17.2                                                                                | n/a                                                                                 | ✓                                                                                 | 50k                                                                                | 46.6                                                                                | n/a                                                                                 | x                                                                                 |                                                                                    | 140                                                                                 | ✓                                                                                   | 70k 85.3 675     |  |  |  |
| 2               | ✓                                                                                 | 40k                                                                                | 15.                                                                                 | n/a                                                                                 | ✓                                                                                 | 30k                                                                                | 14.3                                                                                | n/a                                                                                 | ✓                                                                                 | 55k                                                                                | 34.7                                                                                | n/a                                                                                 | ✓                                                                                 | 50k                                                                                | 50.7                                                                                | n/a                                                                                 | ✓ 10k 78. n/a    |  |  |  |
| 3               | ✓                                                                                 | 60k                                                                                | 15.                                                                                 | n/a                                                                                 | ✓                                                                                 | 45k                                                                                | 15.6                                                                                | n/a                                                                                 | ✓                                                                                 | 90k                                                                                | 47.4                                                                                | n/a                                                                                 | ✓                                                                                 | 20k                                                                                | 52.                                                                                 | n/a                                                                                 | x 135            |  |  |  |
| Avg             | 4/4                                                                               | 34k                                                                                | 14.3                                                                                | 0/4                                                                                 | 4/4                                                                               | 58k                                                                                | 20.                                                                                 | 0/4                                                                                 | 4/4                                                                               | 85k                                                                                | 44.5                                                                                | 0/4                                                                                 | 3/4                                                                               | 58k                                                                                | 58.4                                                                                | 1/4                                                                                 | 2/4 40k 81.7 3/4 |  |  |  |
| Compass Off (R) |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                  |  |  |  |
| Seed            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                  |  |  |  |
| 0               | x                                                                                 |                                                                                    |                                                                                     | n/a                                                                                 | x                                                                                 |                                                                                    |                                                                                     | 945k                                                                                | x                                                                                 |                                                                                    |                                                                                     | 185k                                                                                | x                                                                                 |                                                                                    |                                                                                     | 65k                                                                                 | x 45k            |  |  |  |
| 1               | ✓                                                                                 | 275k                                                                               | 12.4                                                                                | n/a                                                                                 | x                                                                                 |                                                                                    |                                                                                     | 1085k                                                                               | x                                                                                 |                                                                                    |                                                                                     | 110k                                                                                | x                                                                                 |                                                                                    |                                                                                     | 55k                                                                                 | x 40             |  |  |  |
| 2               | x                                                                                 |                                                                                    |                                                                                     | n/a                                                                                 | x                                                                                 |                                                                                    |                                                                                     | 1070k                                                                               | x                                                                                 |                                                                                    |                                                                                     | 165k                                                                                | x                                                                                 |                                                                                    |                                                                                     | 55k                                                                                 | x 25             |  |  |  |
| 3               | x                                                                                 |                                                                                    |                                                                                     | n/a                                                                                 | x                                                                                 |                                                                                    |                                                                                     | 280k                                                                                | x                                                                                 |                                                                                    |                                                                                     | 125k                                                                                | x                                                                                 |                                                                                    |                                                                                     | 90k                                                                                 | x 40             |  |  |  |
| Avg             | 1/4                                                                               | 275k                                                                               | 12.4                                                                                | 0/4                                                                                 | 0/4                                                                               |                                                                                    |                                                                                     | n/a                                                                                 | 0/4                                                                               |                                                                                    |                                                                                     | 4/4                                                                                 | 0/4                                                                               |                                                                                    |                                                                                     | 4/4                                                                                 | 0/4 4/4          |  |  |  |

Table 4: Seed-level terrain-general mining data. (RC) agents are compared to (R) agents in **Island**, **Lake**, **Isthmus**, and **Channel** worlds at 512×512 with initial population 8192.

|                 | Island |      |      |      | Lake |     |      |       | Isthmus |     |      |      | Channel |     |      |      |
|-----------------|--------|------|------|------|------|-----|------|-------|---------|-----|------|------|---------|-----|------|------|
| Compass On (RC) |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| Seed            |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| 0               | ✓      | 40k  | 94.3 | n/a  | x    |     |      | 45k   | ✓       | 45k | 33.  | n/a  | ✓       | 50k | 27.4 | n/a  |
| 1               | ✓      | 30k  | 94.3 | n/a  | ✓    | 30k | 22.5 | n/a   | ✓       | 55k | 37.3 | n/a  | ✓       | 80k | 30.8 | n/a  |
| 2               | ✓      | 30k  | 94.8 | n/a  | x    |     |      | 45k   | ✓       | 40k | 28.1 | n/a  | ✓       | 50k | 41.  | n/a  |
| 3               | ✓      | 70k  | 95.1 | n/a  | ✓    | 35k | 25.3 | n/a   | ✓       | 30k | 41.5 | n/a  | x       |     |      | 80k  |
| Avg             | 4/4    | 43k  | 94.6 | 0/4  | 2/4  | 33k | 23.9 | 2/4   | 4/4     | 43k | 35.  | 0/4  | 3/4     | 60k | 33.1 | 1/4  |
| Compass Off (R) |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| Seed            |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| 0               | x      |      |      | 185k | x    |     |      | 125k  | x       |     |      | 145k | x       |     |      | 100k |
| 1               | ✓      | 75k  | 30.9 | n/a  | x    |     |      | 130k  | x       |     |      | 260k | x       |     |      | 105k |
| 2               | ✓      | 85k  | 47.  | n/a  | ✓    | 75k | 23.4 | 1400k | ✓       | 95k | 18.8 | n/a  | x       |     |      | 105k |
| 3               | ✓      | 115k | 39.5 | n/a  | ✓    | 70k | 24.3 | 1250k | x       |     |      | 140k | x       |     |      | 100k |
| Avg             | 3/4    | 92k  | 39.1 | 1/4  | 2/4  | 73k | 23.9 | 4/4   | 1/4     | 95k | 18.8 | 3/4  | 0/4     |     |      | 4/4  |

Table 5: Seed-level population and foraging data for the vision contribution in the **Ocean** world across scales. 🧑 denotes population size, ⚙️ denotes the percentage of total biomass currently inside agents’ stomachs, 🚶 denotes move actions per agent, and 🍴 denotes eat actions per agent. Reported 🧑, ⚙️, 🚶, and 🍴 values are step averages for each seed.

|                | 1024x1024                                                                           |                                                                                     |                                                                                     |                                                                                     | 512x512                                                                             |                                                                                     |                                                                                     |                                                                                     | 256x256                                                                             |                                                                                     |                                                                                      |                                                                                       | 128x128                                                                               |                                                                                       |                                                                                       |                                                                                       |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                | 32768                                                                               |                                                                                     |                                                                                     |                                                                                     | 8192                                                                                |                                                                                     |                                                                                     |                                                                                     | 2048                                                                                |                                                                                     |                                                                                      |                                                                                       | 512                                                                                   |                                                                                       |                                                                                       |                                                                                       |
| Vision (RCV+A) |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0              | 58354                                                                               | .396                                                                                | .402                                                                                | .426                                                                                | 16413                                                                               | .369                                                                                | .257                                                                                | .532                                                                                | 4195                                                                                | .382                                                                                | .255                                                                                 | .536                                                                                  | 1126                                                                                  | .426                                                                                  | .261                                                                                  | .541                                                                                  |
| 1              | 65107                                                                               | .365                                                                                | .257                                                                                | .529                                                                                | 16200                                                                               | .364                                                                                | .256                                                                                | .533                                                                                | 4142                                                                                | .377                                                                                | .258                                                                                 | .536                                                                                  | 1103                                                                                  | .415                                                                                  | .262                                                                                  | .537                                                                                  |
| 2              | 64735                                                                               | .363                                                                                | .257                                                                                | .530                                                                                | 16560                                                                               | .372                                                                                | .257                                                                                | .533                                                                                | 4269                                                                                | .388                                                                                | .260                                                                                 | .536                                                                                  | 1041                                                                                  | .392                                                                                  | .254                                                                                  | .542                                                                                  |
| 3              | 64595                                                                               | .362                                                                                | .256                                                                                | .532                                                                                | 16206                                                                               | .365                                                                                | .255                                                                                | .531                                                                                | 4249                                                                                | .388                                                                                | .259                                                                                 | .533                                                                                  | 1100                                                                                  | .414                                                                                  | .257                                                                                  | .538                                                                                  |
| Avg            | 63198                                                                               | .371                                                                                | .293                                                                                | .504                                                                                | 16345                                                                               | .368                                                                                | .256                                                                                | .532                                                                                | 4213                                                                                | .384                                                                                | .258                                                                                 | .535                                                                                  | 1092                                                                                  | .412                                                                                  | .259                                                                                  | .539                                                                                  |
| Blind (RC+A)   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0              | 22734                                                                               | .143                                                                                | .446                                                                                | .372                                                                                | 6549                                                                                | .173                                                                                | .442                                                                                | .400                                                                                | 2036                                                                                | .209                                                                                | .433                                                                                 | .461                                                                                  | 545                                                                                   | .219                                                                                  | .465                                                                                  | .427                                                                                  |
| 1              | 23670                                                                               | .151                                                                                | .426                                                                                | .388                                                                                | 6939                                                                                | .180                                                                                | .438                                                                                | .396                                                                                | 3107                                                                                | .321                                                                                | .390                                                                                 | .416                                                                                  | 1082                                                                                  | .444                                                                                  | .416                                                                                  | .460                                                                                  |
| 2              | 24745                                                                               | .159                                                                                | .358                                                                                | .426                                                                                | 5448                                                                                | .136                                                                                | .474                                                                                | .374                                                                                | 1587                                                                                | .163                                                                                | .440                                                                                 | .433                                                                                  | 979                                                                                   | .421                                                                                  | .404                                                                                  | .436                                                                                  |
| 3              | 20004                                                                               | .130                                                                                | .469                                                                                | .359                                                                                | 10354                                                                               | .265                                                                                | .311                                                                                | .439                                                                                | 3164                                                                                | .324                                                                                | .321                                                                                 | .450                                                                                  | 788                                                                                   | .317                                                                                  | .431                                                                                  | .477                                                                                  |
| Avg            | 22788                                                                               | .146                                                                                | .425                                                                                | .386                                                                                | 7323                                                                                | .189                                                                                | .416                                                                                | .402                                                                                | 2473                                                                                | .254                                                                                | .396                                                                                 | .440                                                                                  | 848                                                                                   | .351                                                                                  | .429                                                                                  | .450                                                                                  |



Table 6: Seed-level predation data for the vision contribution in the **Ocean** world across scales.  denotes attacks per agent, averaged across agents as the fraction of time an agent attacked.  denotes homicides per attack. Reported  and  values are step averages for each seed. The 0.562  outlier for seed 3 in the 128x128 (**RC+A**) run occurs because that seed went extinct before 2M steps, causing attacks per agent to spike as population size dwindled.

|                | 1024x1024                                                                           |                                                                                     | 512x512                                                                             |                                                                                     | 256x256                                                                             |                                                                                     | 128x128                                                                             |                                                                                       |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                | 32768                                                                               |                                                                                     | 8192                                                                                |                                                                                     | 2048                                                                                |                                                                                     | 512                                                                                 |                                                                                       |
| Vision (RCV+A) |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |
| 0              | .011                                                                                | .776                                                                                | .015                                                                                | .731                                                                                | .013                                                                                | .784                                                                                | .010                                                                                | .788                                                                                  |
| 1              | .016                                                                                | .733                                                                                | .015                                                                                | .741                                                                                | .014                                                                                | .780                                                                                | .011                                                                                | .793                                                                                  |
| 2              | .016                                                                                | .707                                                                                | .015                                                                                | .737                                                                                | .014                                                                                | .750                                                                                | .012                                                                                | .771                                                                                  |
| 3              | .016                                                                                | .712                                                                                | .015                                                                                | .780                                                                                | .013                                                                                | .759                                                                                | .011                                                                                | .765                                                                                  |
| Avg            | .015                                                                                | .732                                                                                | .015                                                                                | .747                                                                                | .014                                                                                | .758                                                                                | .011                                                                                | .779                                                                                  |
| Blind (RC+A)   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |
| 0              | .070                                                                                | .218                                                                                | .042                                                                                | .437                                                                                | .005                                                                                | 1.074                                                                               | .016                                                                                | .558                                                                                  |
| 1              | .054                                                                                | .300                                                                                | .044                                                                                | .383                                                                                | .016                                                                                | .871                                                                                | .006                                                                                | 1.032                                                                                 |
| 2              | .076                                                                                | .236                                                                                | .048                                                                                | .238                                                                                | .031                                                                                | .383                                                                                | .010                                                                                | .979                                                                                  |
| 3              | .059                                                                                | .203                                                                                | .044                                                                                | .456                                                                                | .033                                                                                | .446                                                                                | .562                                                                                | .841                                                                                  |
| Avg            | .065                                                                                | .239                                                                                | .045                                                                                | .378                                                                                | .021                                                                                | .693                                                                                | .148                                                                                | .853                                                                                  |

## B DEMOGRAPHY

This appendix provides a demographic view of the paper’s scale and compass contributions. We compare extinction and mining-emergence rates as a function of world size and access to global direction through the compass sensor. Extinction means that the population reaches zero individuals, and mining is defined as in the main experiments: a 10% drop in free biomass on land.

Runs are grouped based on both world size and compass access. We simulated 64 runs each of  $128 \times 128$  with no compass and  $128 \times 128$  with compass, and 32 runs each of  $512 \times 512$  with no compass and  $512 \times 512$  with compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution (treating runs that did not go extinct within 2M steps as right-censored). In Figures 7 and 8 we observe that worlds tend to survive longer and produce mining more quickly if they are large or if agents have compass information.

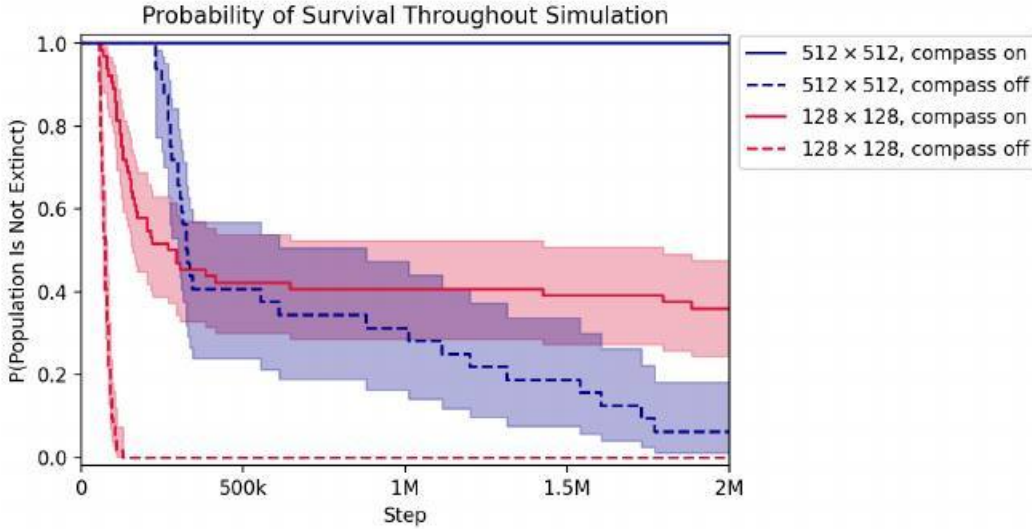


Figure 7: Probability the population has not gone extinct given a world size, compass access, and timestep. We fit a unique non-parametric Kaplan–Meier estimator for each group of runs (e.g.  $512 \times 512$ , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each  $512 \times 512$  group fits the estimator on 32 runs; each  $128 \times 128$  group fits on 64 runs.

We compare emergence and extinction timing with a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazard rates (Table 7). Runs that did not experience an emergence or extinction event by the end of 2M time steps were treated as censored. For extinction, both larger world size and compass use substantially reduce the hazard; the extinction rate in  $128 \times 128$  is about 25 times greater than the corresponding rate in  $512 \times 512$  worlds ( $HR=0.04$ ). Enabling the compass produces an effect of a similar magnitude ( $HR=0.03$ ). For mining emergence, larger worlds generate mining behavior much more rapidly ( $HR=6.07$ ), and compass use accelerates emergence dramatically ( $HR=34.59$ ). The models show good discriminative ability between groups (concordance 0.80 for extinction, 0.85 for mining). We find that larger worlds can sustain larger populations, reducing the chance of hitting the absorbing zero boundary while simultaneously exploring a larger phenotype space due to greater quantities of mutation as compared to smaller worlds.



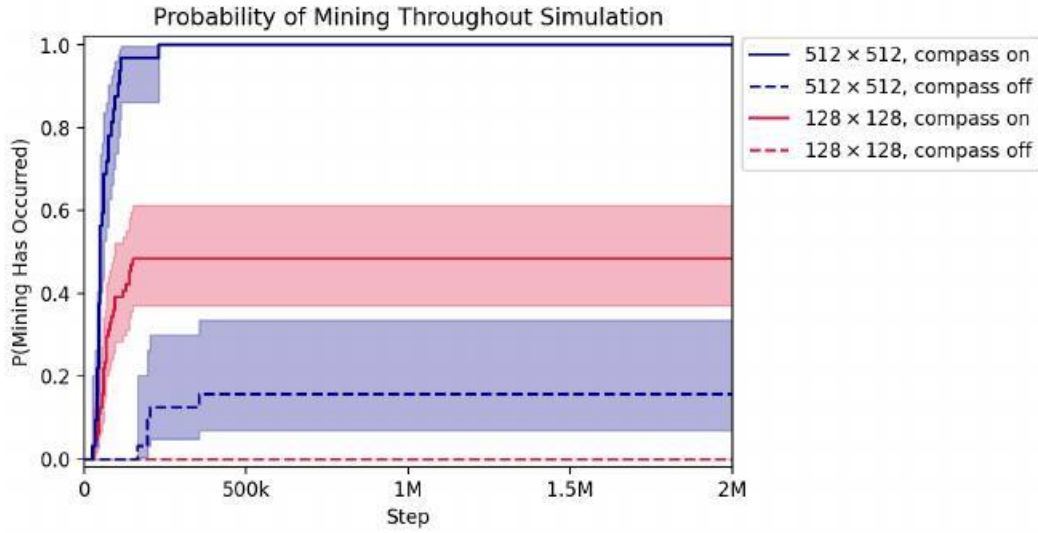


Figure 8: Probability mining has evolved within the population given a world size, compass access, and timestep. We fit a unique non-parametric Kaplan-Meier estimator for each group of runs (e.g.  $512 \times 512$ , compass on). The shaded areas represent the 95% confidence intervals for each curve. Each  $512 \times 512$  group fits the estimator on 32 runs; each  $128 \times 128$  group fits on 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events with a  $128 \times 128$ , compass off baseline. Hazard ratios (HR) less than 1 indicate reduced hazard (longer time to event); for example, extinction  $HR = 0.04$  for  $512 \times 512$  means extinction occurs at 0.04 times the rate of  $128 \times 128$ . There are 192 total runs: 32 seeds with compass on and 32 seeds with compass off in the  $512 \times 512$  world, and 64 seeds with compass on and 64 seeds with compass off in the  $128 \times 128$  world.

| Covariate             | Extinction |           |           | Mining   |             |           |
|-----------------------|------------|-----------|-----------|----------|-------------|-----------|
|                       | HR         | 95% CI    | p-value   | HR       | 95% CI      | p-value   |
| Size $512 \times 512$ | 0.04       | 0.02–0.07 | $< 0.005$ | 6.07     | 3.61–10.20  | $< 0.005$ |
| Compass On            | 0.03       | 0.01–0.05 | $< 0.005$ | 34.59    | 13.51–88.57 | $< 0.005$ |
| <b>Concordance</b>    | 0.80       |           |           | 0.85     |             |           |
| <b>Events / Total</b> | 135 / 192  |           |           | 68 / 192 |             |           |

## C ADDITIONAL FORAGING AND PREDATION RESULTS

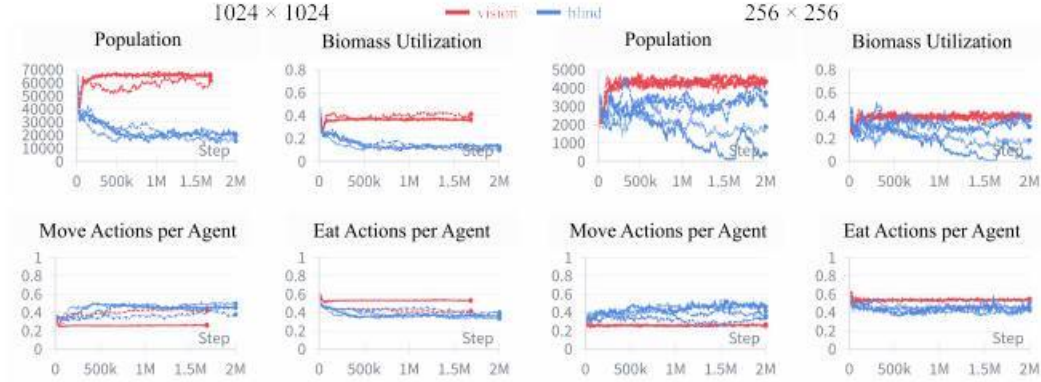


Figure 9: Additional scale comparison for the foraging contribution in the **Ocean** world. Vision agents (red) produce more stable populations and greater biomass utilization through more frequent effective eating and less movement. The larger-scale runs show lower variance across seeds. Because of GPU-hour limits, the vision seeds at  $1024 \times 1024$  ran for 1.7M rather than 2M steps.

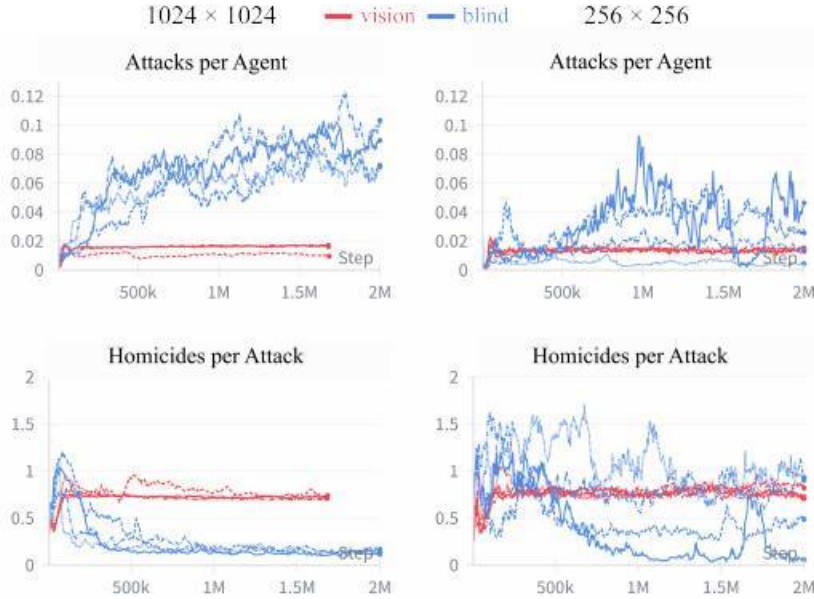


Figure 10: Additional scale comparison for the predation contribution in the **Ocean** world. With vision (red), agents attack less frequently but with much higher precision, suggesting specialized predation alongside the foraging improvements in Figure 9. The effect is more pronounced at larger scale, while blind agents (blue) show greater behavioral variance in smaller worlds.

## D ADDITIONAL EXPERIMENT VISUALIZATIONS



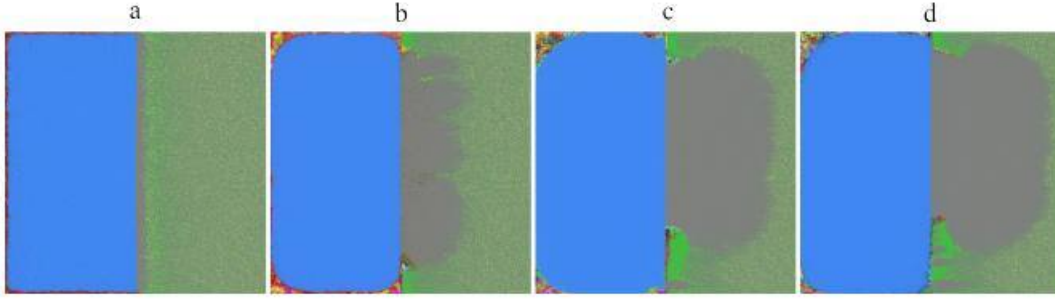


Figure 11: Large-scale **Beach** visualization showing why mining is an ecological contribution rather than a single-agent pathfinding event. In this  $1024 \times 1024$  world, **(RC)** agents first die near the shore and transfer biomass to dry land (a), then evolve mining that clears the beach (b). When biomass later re-accumulates at the beach ends (c), subpopulations re-develop mining and travel inland along smaller biomass deposits to extract those resources (d).

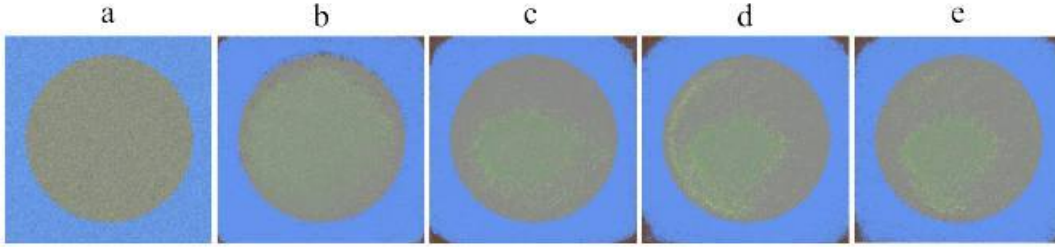


Figure 12: Terrain-general mining behavior in a  $512 \times 512$  **Island** world with **(RC)** agents. Different shore regions are mined over the run (b) (c); when mining becomes difficult, agents die before returning to water and transfer biomass back to the island (d). Later lineages adapt to re-mine this accumulated wall of biomass and clear it from the shore (e).

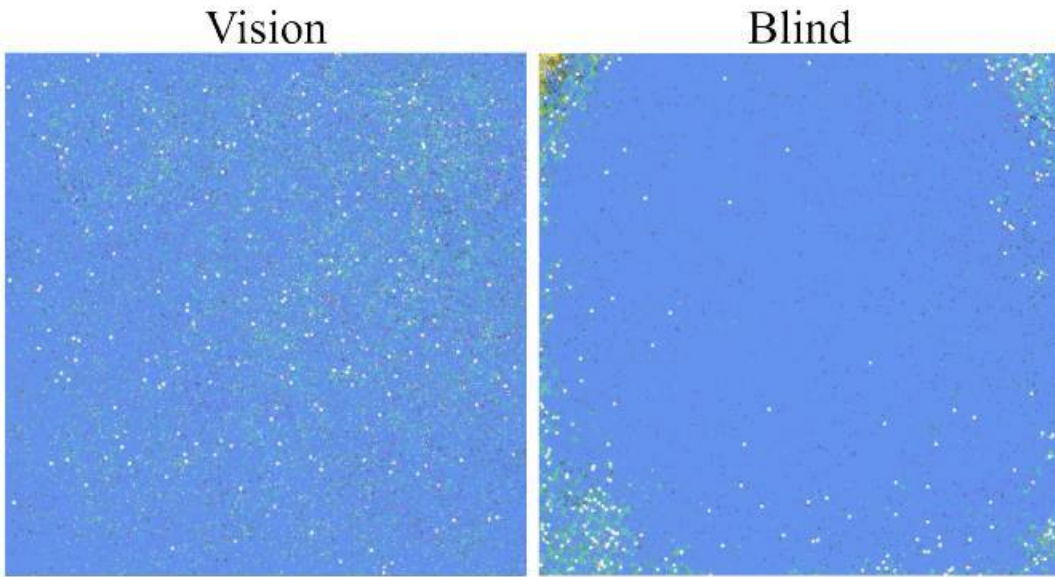


Figure 13: Representative visual comparison supporting the foraging and predation contribution in a  $512 \times 512$  **Ocean** world with attacks enabled. Vision agents **(RCV+A)** are shown on the left and blind agents **(RC+A)** on the right; white squares mark attack locations. With vision, agents locate free resources throughout the ocean and prey more effectively (Figures 5 and 6), while blind agents explore less and instead adapt corner-clustering policies.

## E DETAILS ON COMPUTATION AND REPRODUCIBILITY

This appendix makes the computational contribution explicit by detailing the simulator design, runtime scaling, and reproducibility considerations behind the main ecological experiments.

**Simulator design.** We conduct all experiments using a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. Our simulator computes every part of the environment step on the GPU, including terrain dynamics, resource flows, agent updates, and measurement, so that policy evaluation and simulation share a single JAX compilation pipeline. The simulator exposes a minimal population-level interface: given current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation allows the same environment implementation to be reused with different evolutionary algorithms, reinforcement learning methods, or other population update rules. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This yields the throughput required for our scaling experiments, with runtime measurements for a single H100 GPU given in Table 8.

A key feature of the simulator is its built-in tooling, including a measurement API for computing and recording aggregate statistics such as biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters at each step, which we use directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from generated reports, and supports inspection of individual agents (Figure 1). These tools were critical for discovering and validating qualitative phenomena such as long-range resource mining and predation patterns that are not obvious from scalar metrics alone.

From a high-performance computing perspective, our simulator fills a gap relative to existing simulators. General-purpose agent-based frameworks such as MASON (Luke et al. 2005), FLAME-GPU (Richmond et al. 2024), or Griddly (Bamford et al. 2020) offer high configurability but require substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study. Conversely, GPU-native multi-agent RL environments emphasize fixed numbers of agents and task-specific reward structures. We combine GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over  $10^5$  agents while remaining interoperable with diverse population algorithms. This combination yields a state-of-the-art platform for scaling ecological experiments like those in this paper and a useful foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

**Runtime data.** Table 8 shows runtime data across different grid sizes and maximum populations of agents. Each run was conducted on a single H100 GPU for a maximum of 2M steps. Note that while this table shows run-times for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ( $\sim 22k$  parameters total), the runtime varies with the size of the agent networks, the available sensors, and the level of logging.

Our largest runs at the  $1024 \times 1024$  grid size take  $\sim 10$  hours in the **Beach** terrain and  $\sim 14$  hours in the **Ocean** terrain on one H100, with memory usage of roughly 60GB. **Ocean** runs are slightly slower due to the enabled agent attack action and vision sensing.

**Reproducibility.** Due to the large and randomized nature of our environments, there can be substantial differences in the precise outcomes of each experiment from run to run. As a result, we report the results of four different random seeds for each experiment. Unfortunately, due to nondeterminism in some low-level GPU operations, we find that repeating an experiment with the same random seed does not reliably produce the same results. In particular, modern GPUs may execute parallel floating-point operations such as summations in different orders across runs and fuse operations in ways that change rounding behavior. Very small differences arising from finite-precision arithmetic can accumulate over many steps into noticeable drift in the simulation trajectory. The use of 16-bit weights can coarsen these rounding effects, but is not the primary source. Despite differences across runs, clear high-level patterns emerge when the results of multiple seeds are taken in aggregate.



Table 8: Runtime data in the **Beach** and **Ocean** worlds across different grid sizes and maximum populations of agents. The **(RC)** agents each have a 22,465 parameter neural network. The **(RCV + A)** agents each have a 22,529 parameter neural network. Simulation steps per second (hz) and total extrapolated wall-clock runtime in hours for 2M steps are reported. Effective runtime is less for seeds in which populations go extinct before 2M steps.

|                 | 1024 × 1024 |          | 512 × 512 |         | 256 × 256 |         | 128 × 128 |         | 64 × 64 |         |
|-----------------|-------------|----------|-----------|---------|-----------|---------|-----------|---------|---------|---------|
|                 | 131072      |          | 32768     |         | 16384     |         | 8192      |         | 4096    |         |
| Environment     | hz          | runtime  | hz        | runtime | hz        | runtime | hz        | runtime | hz      | runtime |
| Beach (RC)      | 55          | 10.1 hrs | 190       | 2.9 hrs | 315       | 1.8 hrs | 535       | 1.0 hrs | 750     | 0.7 hrs |
| Ocean (RCV + A) | 40          | 13.9 hrs | 142       | 3.9 hrs | 235       | 2.4 hrs | 415       | 1.3 hrs | 750     | 0.7 hrs |

## F ABLATIONS

These ablations support the main contribution claims by testing whether scale- and sensing-dependent behaviors persist under changes to population initialization, resource rules, mutation, network architecture, vision, attack strength, and agent health.

### F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

This ablation separates the contribution of world scale from the possible confound of initial population size. In a fixed 256×256 **Beach** world, we vary the initial population and measure mining consistency and population stability for compass agents **(RC)**. Agents without a compass **(R)** provide a negative control: they show 0% mining and 100% extinction at every initial population size except 8192, where they show 25% mining and 75% extinction. Mining and extinction frequencies need not sum to 100% because a population can mine and later go extinct, or survive for 2M steps without mining.

Figure 14 shows that initial population size does not explain the main mining result in this setting. Populations initialized with only 128 agents evolve mining as frequently as populations initialized with 8192 agents, and extinction rates remain largely stable across the sweep.

Figure 15 explains why initial population size has limited effect. Regardless of initialization, runs quickly approach a similar population size during the first 50,000 steps. The fixed resource base therefore induces an implicit carrying capacity: populations below it grow toward it, while populations above it are constrained by it. After this early correction, effective population size is similar enough that mining and extinction frequencies are largely unchanged. Larger initial populations still add some resources because agents deposit carried biomass on death, which raises maximum population size on average (Table 9); however, this shift is not large enough to change mining or extinction frequency in this setting. Later population trajectories can still diverge substantially as different emergent abilities reshape resource distributions at different times.

### F.2 VARYING RESOURCE RULES

This ablation tests whether the mining contribution depends on the resource regime. Using the same 256×256 **Beach** setup as Appendix F.1, we fix the initial population at 2048 compass agents **(RC)** and vary biomass photosynthesis rate and initial biomass site density.

Biomass photosynthesis controls how quickly landscape biomass creates energy over time, while initial biomass site density controls the fraction of grid cells containing biomass at the start of simulation. The main experiments in Section 4 use biomass photosynthesis rate 0.01 and initial biomass site density 1/4.

Unlike initial population size (Figure 14), resource rules substantially affect mining and extinction consistency (Figures 16 and 17).

The mechanism is consistent with the scale argument in the main paper. Larger photosynthesis rates or initial biomass densities raise the effective carrying capacity by increasing available resources.

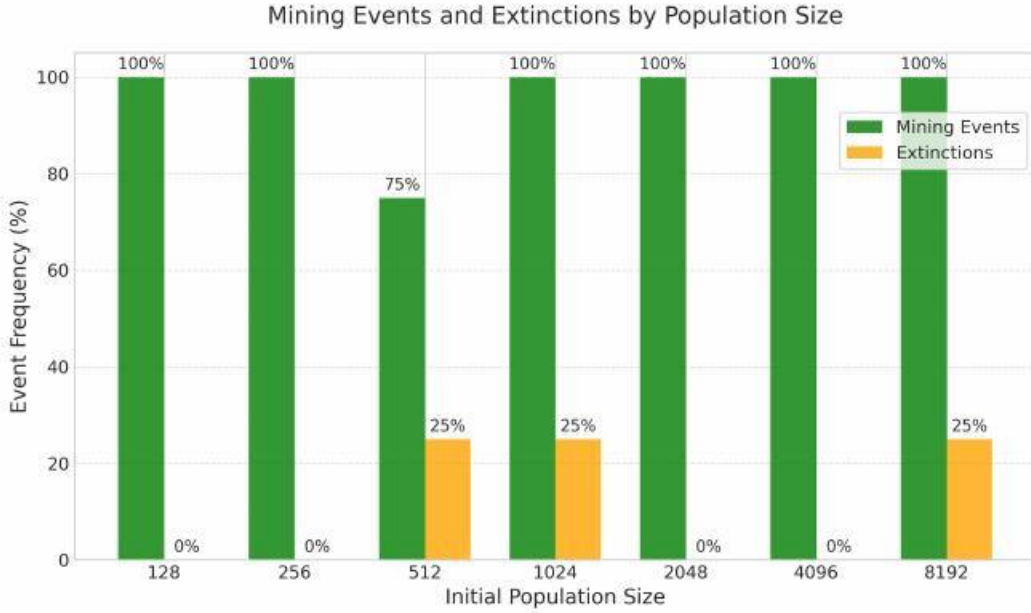


Figure 14: Initial-population ablation for the mining contribution. The plot reports mining and extinction frequencies in a  $256 \times 256$  **Beach** world for (RC) agents, with four random seeds per initial population size. A 25% Extinction Event Frequency means that 1/4 seeds reached population 0 before 2M simulation steps.

Table 9: Maximum post-initialization population size in the initial-population ablation, reported as averages and standard deviations across four random seeds. Larger initial populations reach slightly higher maxima and show greater variance, but the maximum is mostly robust: an initial population of 128, only 1.6% of 8192, still reaches 78.9% of the latter’s maximum population (3178 vs. 4026).

| Initial Population Size | Max Population Size After Step 0 |
|-------------------------|----------------------------------|
| 128                     | $3178 \pm 217$                   |
| 256                     | $3056 \pm 61$                    |
| 512                     | $3451 \pm 437$                   |
| 1024                    | $3613 \pm 381$                   |
| 2048                    | $3786 \pm 431$                   |
| 4096                    | $3689 \pm 437$                   |
| 8192                    | $4026 \pm 642$                   |

Larger sustained populations create more opportunities for mutations that support mining while reducing extinction risk. The data match this interpretation: excluding extinct seeds, mean maximum population after step 0 increases from 1608 at initial biomass site density 1/16 to 2466 at 1/8, 3786 at 1/4, and 5353 at 1/2.

### F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

This ablation addresses a key alternative explanation for the scale contribution: the increased stability of mining in large environments (Table I) might reflect self-averaging rather than qualitatively different ecological dynamics. By self-averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as  $1/\sqrt{N}$ , where  $N$  is the system size, such that a single large system behaves like the average of smaller systems of the same total size. Under this view, a  $512 \times 512$  **Beach** world might simply be acting like many loosely cou-



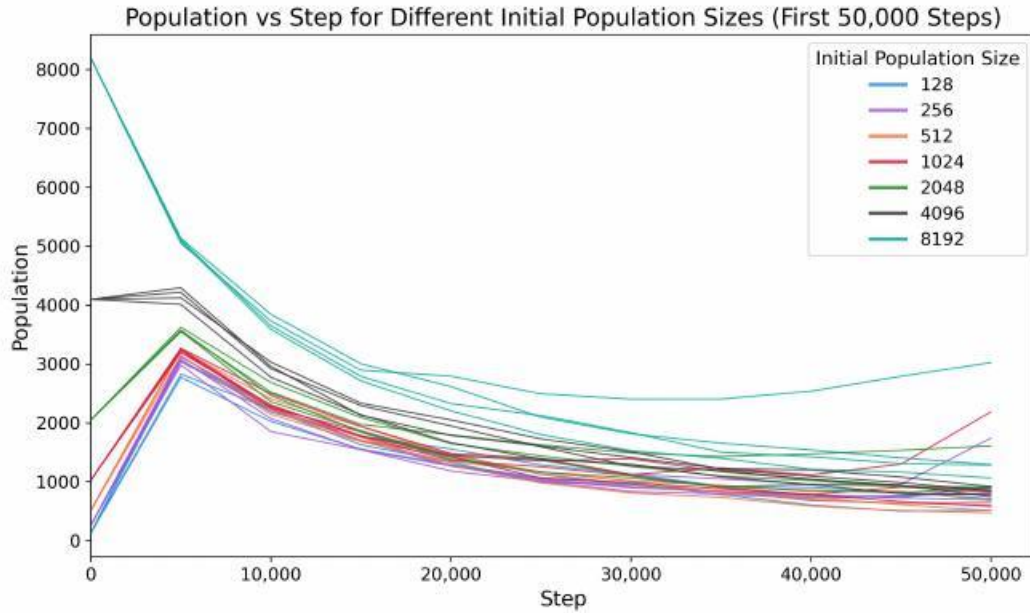


Figure 15: Early carrying-capacity dynamics in the initial-population ablation. Population size is shown for the first 50,000 steps in a  $256 \times 256$  **Beach** world with (RC) agents and four seeds per initial population size. All initial populations approach a similar resource-determined carrying capacity early in the run, while larger initial populations raise that capacity slightly through biomass deposited on death.

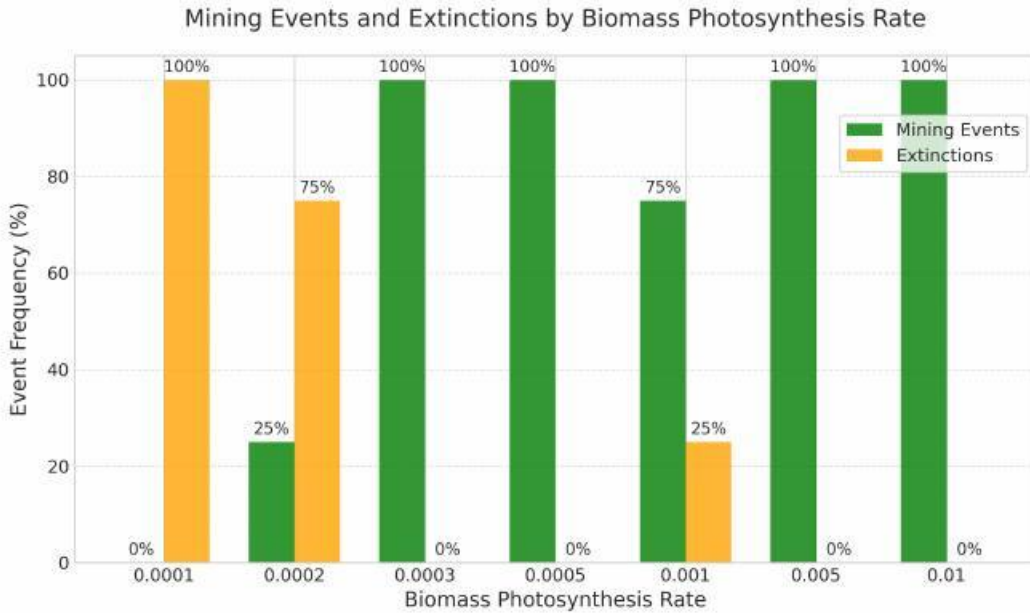


Figure 16: Resource-rule ablation for biomass photosynthesis rate in a  $256 \times 256$  **Beach** world. Mining and extinction frequencies are computed across four random seeds per setting. Mining emerges consistently at biomass photosynthesis rates of 0.0003 and greater.

pled  $128 \times 128$  worlds run in parallel. If that were the case, then our scaling results could in principle be reproduced by running enough small worlds and averaging over random seeds.

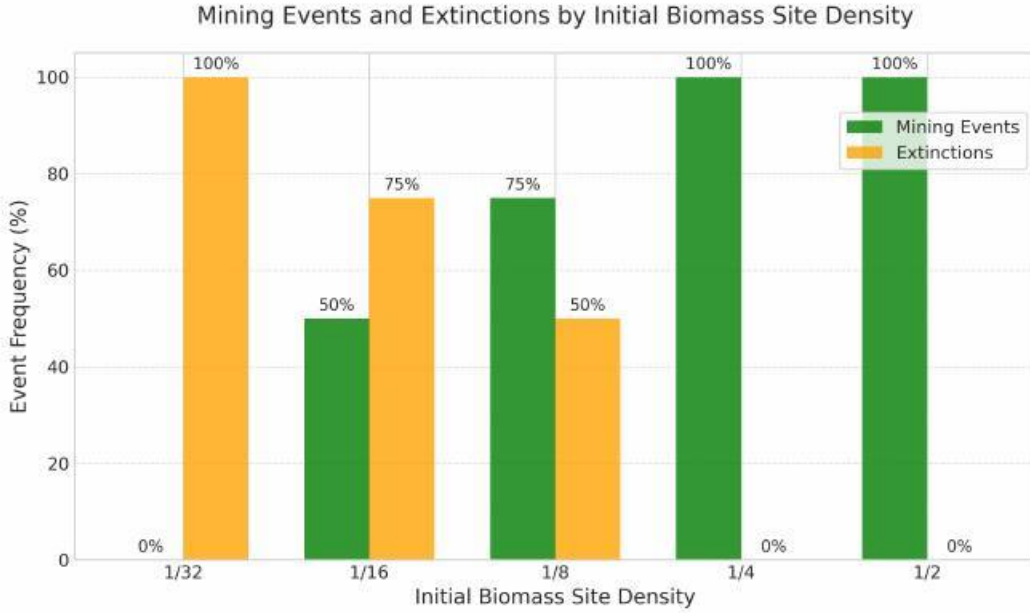


Figure 17: Resource-rule ablation for initial biomass site density in a  $256 \times 256$  **Beach** world. Mining and extinction frequencies are computed across four random seeds per setting. Higher initial biomass site density makes mining more consistent and extinction less likely.

To test this hypothesis, we hold total “sample size” roughly fixed while changing how cells, resources, and agents are partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a  $512 \times 512$  world with 8192 initial agents each,
- 16 runs of a  $256 \times 256$  world with 2048 initial agents each,
- 64 runs of a  $128 \times 128$  world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1. As a control, we repeat this protocol for agents without a compass (**R**). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

If large environments were acting as self-averages of many smaller ones, we would expect mining and extinction rates to be roughly invariant to how we distribute grid cells and agents across runs: 4 large worlds or 64 small ones would exhibit similar probabilities of discovering mining. Instead, we observe a strong dependence on world size. For (**RC**) agents, mining emerges in all runs with no extinctions at  $512 \times 512$ , remains common but not guaranteed at  $256 \times 256$ , and becomes unreliable and fragile at  $128 \times 128$ , where many populations go extinct before reaching 2M steps. In contrast, (**R**) agents never develop mining and go extinct at all scales, confirming that the compass remains a key enabler of long-range behavior.

These results are inconsistent with a self-averaging explanation. On the contrary, larger environments appear to enable and stabilize mining behavior in ways that cannot be explained solely by averaging over more small simulations.

Even when matching the total number of cells, agents, and simulation steps across conditions, concentrating resources into a few large worlds produces a regime in which mining is consistent and extinctions are rare, while dispersing the same resources across many small worlds gives populations



Table 10: Self-averaging check for the scale contribution: mining and extinction events when trading off world size against the number of independent runs. We compare **Beach** worlds at grid sizes  $512 \times 512$ ,  $256 \times 256$ ,  $128 \times 128$  with initial populations scaled proportionally (8192, 2048, 512 agents, respectively). For each size, we allocate the same total number of grid cells and initial agents across all runs by running 4 seeds at  $512 \times 512$ , 16 seeds at  $256 \times 256$ , and 64 seeds at  $128 \times 128$ . We report the fraction of runs in which mining emerged (🔑) and the fraction in which the population went extinct before 2M steps (💀), for agents with a compass (**RC**) and without a compass (**R**).

|               | $512 \times 512$ |     | $256 \times 256$ |       | $128 \times 128$ |       |
|---------------|------------------|-----|------------------|-------|------------------|-------|
|               | 8192             |     | 2048             |       | 512              |       |
|               | 🔑                | 💀   | 🔑                | 💀     | 🔑                | 💀     |
| ( <b>RC</b> ) | 4/4              | 0/4 | 15/16            | 1/16  | 31/64            | 41/64 |
| ( <b>R</b> )  | 0/4              | 4/4 | 0/16             | 16/16 | 0/64             | 64/64 |

that frequently fail to discover mining and are more likely to collapse. This suggests that increasing environmental scale is not simply smoothing out stochastic noise, but is altering the underlying ecological dynamics, for example by supporting larger, more spatially distributed populations and more persistent diversity in evolving policies.

#### F.4 MUTATION RATE SWEEP

This ablation asks whether the mining contribution is confined to one mutation-rate setting. The mutation rate controls the scale of Gaussian noise added to a child policy at birth, and therefore controls how different a child is from its parent. We run  $256 \times 256$  **Beach** worlds with an initial population of 2048 (**RC**) agents for up to 2M steps.

Figure 18 shows that mining appears in at least 3 of 4 seeds for mutation rates from  $6 \times 10^{-3}$  to  $7.875 \times 10^{-2}$ , so the Section 4.1 result at  $3 \times 10^{-2}$  is not a narrow hyperparameter artifact. Very small mutation rates, such as  $3 \times 10^{-3}$ , limit behavioral innovation and impair survival. Very large mutation rates, such as  $3 \times 10^{-1}$ , destabilize populations and lead to extinction without mining.

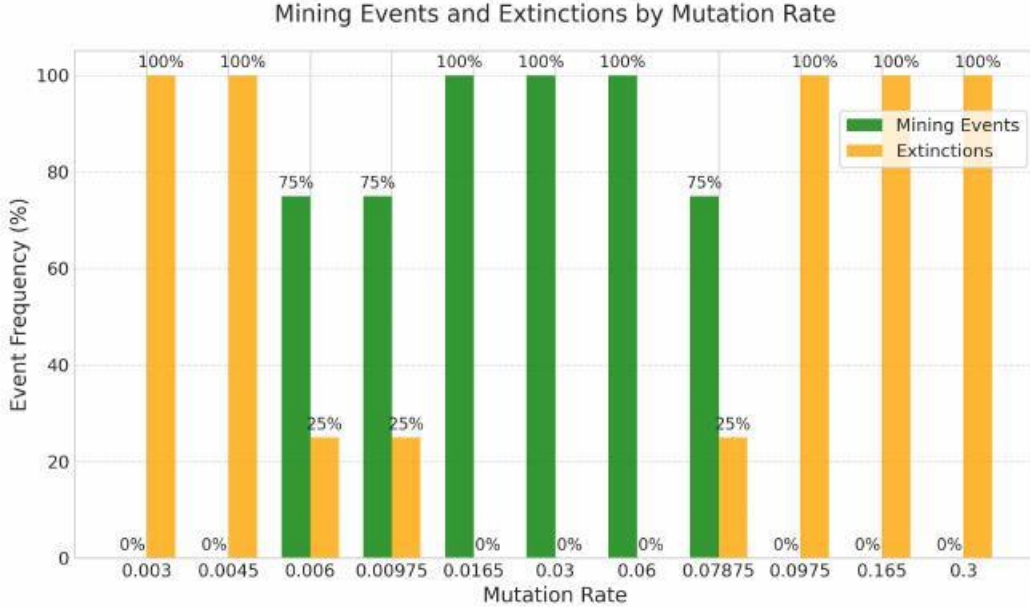


Figure 18: Mutation-rate robustness for the mining contribution. Mining and extinction frequencies are shown for a  $256 \times 256$  **Beach** world, with four random seeds per mutation-rate setting.

## F.5 NETWORK SIZE

This ablation tests whether the scale contribution depends on one policy-network size or architecture. We investigate  $256 \times 256$  and  $512 \times 512$  **Beach** worlds with initial population sizes of 2048 and 8192 (**RC**) agents, respectively (scaling by four with the number of grid cells). We ran baselines with (**R**) agents for each experiment, again finding rare to no mining events and 100% extinction rates.

The main experiments in Section 4 use 2-layer, 64-channel policies with 22465 parameters per agent (Table 11). In Figure 19 we show that in the  $256 \times 256$  **Beach**, halving channels to 32 (resulting in 9185 total parameters) or doubling to 128 (giving 61313 parameters) with number of layers fixed has minimal to no effect on mining and extinction frequency. Meanwhile, changing network depth has a significant effect: all populations of agents with just 1 layer and 32 channels went extinct, and 50% of populations of agents with 3 layers and 128 channels went extinct.

Crucially, scaling the environment to the  $512 \times 512$  grid size produced significantly different dynamics for networks of varying depth (Figure 20). In particular, in the larger world, agent populations with 3 layer, 128 channel networks consistently mined and survived through 2M steps. Surprisingly, agent populations with the small 1 layer, 32 channel networks mined consistently as well. We hypothesize that it is more challenging for both smaller and larger networks to discover new behaviors for different reasons. When the network is smaller, the reduced capacity makes it hard to find good solutions. When the network is larger, the increased size of the search space means it takes longer to find promising new behavior. Regardless of the reasons, the fact that these networks fail so often in small environments but succeed in larger ones further illustrates the stability that comes from increasing scale. The larger environment can support larger populations, which allows for more exploration in policy space. Simultaneously, it takes longer for locally successful mutants to propagate across a large environment, which promotes diversity and again ensures greater exploration. For more discussion of these effects see (Mitchell et al., 2006).

Table 11: Policy-network parameter counts for the network-size ablation, shown for (**RC**) and (**R**) agents across architectures.

| Architecture<br>(layers, channels) | Parameter Count |              |
|------------------------------------|-----------------|--------------|
|                                    | ( <b>RC</b> )   | ( <b>R</b> ) |
| 1, 32                              | 8129            | 7969         |
| 2, 32                              | 9185            | 9025         |
| 2, 64                              | 22465           | 22145        |
| 2, 128                             | 61313           | 60673        |
| 3, 128                             | 77825           | 77185        |

## F.6 VARYING VISION RANGE

This ablation refines the vision contribution by testing how visual range and viewing geometry affect foraging and predation in the  $512 \times 512$  **Ocean** world with (**RCV+A**) agents. We evaluate visual sensor ranges of  $3 \times 3$ ,  $7 \times 7$ , and  $9 \times 9$ . For each range, we evaluate two viewing configurations shown in Figure 21: a) Around: agents can see grid cells all around them b) Front: agents cannot see cells behind them. The total number of cells in the agent’s vision range is equal for the two configurations. The main experiments with vision in Section 4.2 were run with the  $7 \times 7$  Around setting.

These results clarify the visual-sensing contribution: a  $3 \times 3$  range behaves similarly to no vision (Blind), while  $9 \times 9$  performs similarly to  $7 \times 7$ . This indicates that there may be diminishing returns of larger vision ranges in the current environmental configuration. This is intuitive given the rest of the environmental configuration; since the agents have no memory, it is unlikely that visual features at the periphery of a  $9 \times 9$  window could meaningfully alter the optimal action relative to a  $7 \times 7$  window consistently enough to provide a strong learning signal.

The data also suggest that for  $7 \times 7$  and  $9 \times 9$  vision, the Around configuration is slightly more favorable for population stability and effective attack and eat actions, yielding higher average population



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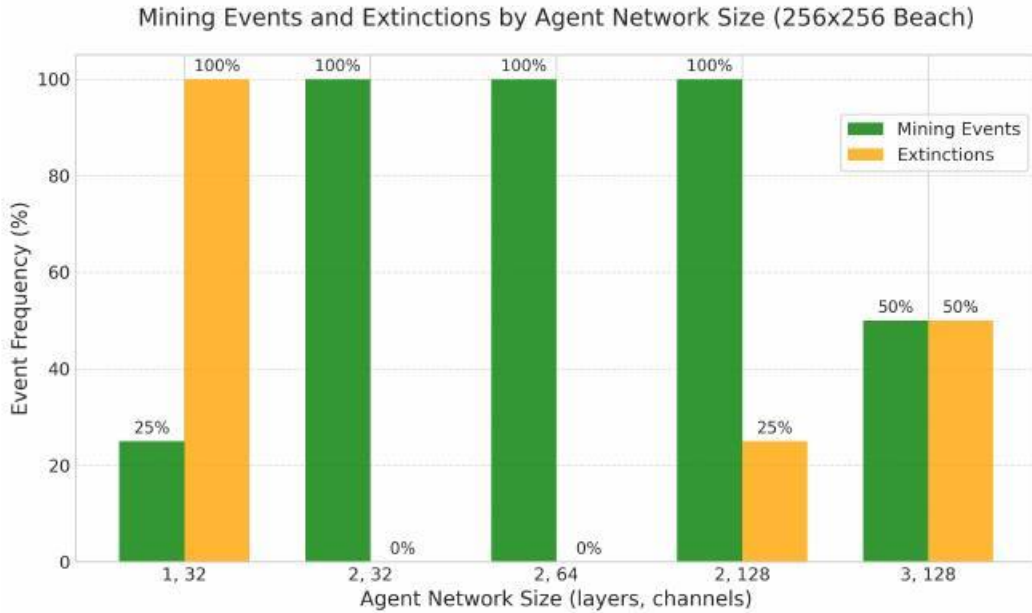


Figure 19: Network-size ablation at  $256 \times 256$  for the mining contribution. Mining and extinction frequencies are computed across four random seeds per policy architecture.

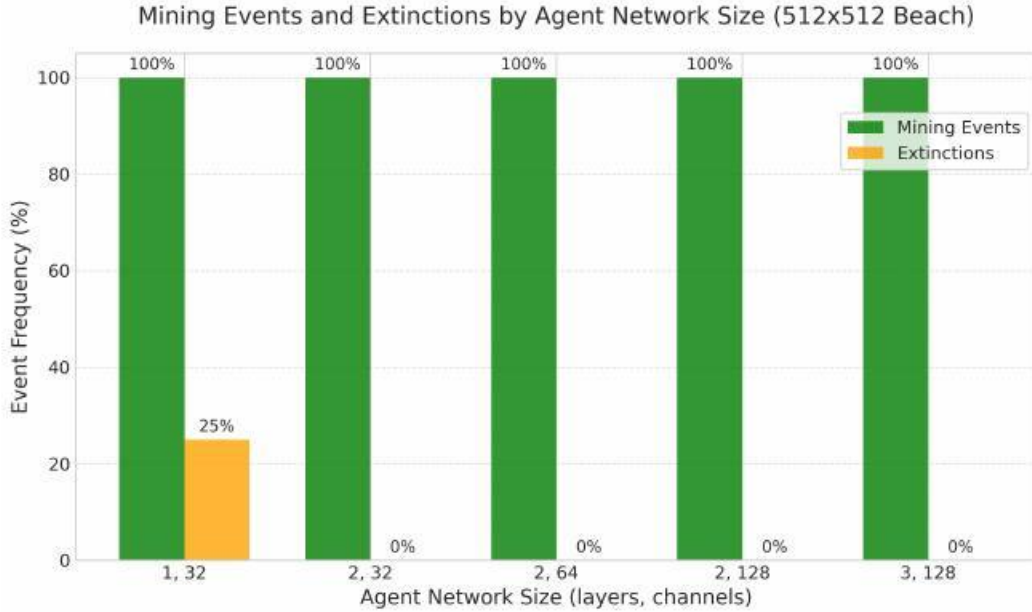


Figure 20: Network-size ablation at  $512 \times 512$  for the mining contribution. Mining and extinction frequencies are computed across four random seeds per policy architecture, showing greater robustness at larger scale.

biomass utilization , and homicides per attack than the Front configuration. We hypothesize that this is due to diminishing returns of being able to see more distant pixels; in the Around configuration, more pixels in the agent's vision range are nearby, likely containing more relevant information for near-term actions than distant cells visible with the Front configuration. At the  $3 \times 3$  vision range, however, the trend is reversed, with the Front configuration being more favorable than Around; this indicates that in this environment, the marginal benefit of seeing one more pixel ahead

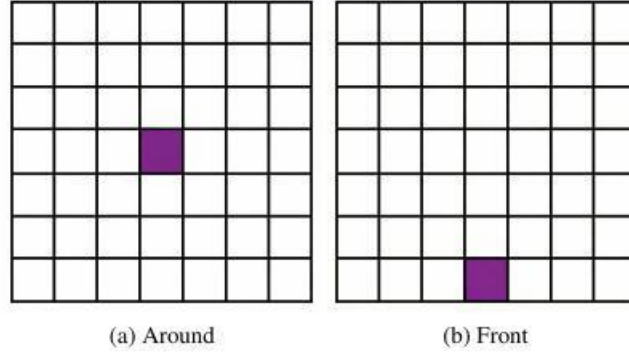


Figure 21: Viewing geometries used in the vision-range ablation for the  $7 \times 7$  sensor. The purple cell is the agent, oriented toward the top of the page.

Table 12: Vision-range ablation for the foraging and predation contribution in the  $512 \times 512$  **Ocean** world with (RCV+A) agents. Refer to Tables 5 and 6 for definitions of the metrics represented by each icon. For each seed, the value for each metric was averaged across all steps starting at step 500k to allow them to stabilize. Reported values are the mean across the four seeds and the standard deviation of the four seed averages.

| Vision Range | Viewing Configuration |  |  |  |  |  |  |
|--------------|-----------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $3 \times 3$ | Blind                 | $7212 \pm 2496$                                                                   | $.181 \pm .060$                                                                   | $.435 \pm .090$                                                                   | $.392 \pm .033$                                                                     | $.032 \pm .006$                                                                     | $.307 \pm .090$                                                                     |
|              | Around                | $10308 \pm 4016$                                                                  | $.258 \pm .105$                                                                   | $.352 \pm .014$                                                                   | $.427 \pm .022$                                                                     | $.040 \pm .030$                                                                     | $.388 \pm .195$                                                                     |
|              | Front                 | $13863 \pm 520$                                                                   | $.351 \pm .023$                                                                   | $.336 \pm .008$                                                                   | $.437 \pm .009$                                                                     | $.026 \pm .006$                                                                     | $.470 \pm .145$                                                                     |
| $7 \times 7$ | Around                | $16739 \pm 122$                                                                   | $.373 \pm .003$                                                                   | $.259 \pm .000$                                                                   | $.532 \pm .001$                                                                     | $.015 \pm .000$                                                                     | $.751 \pm .007$                                                                     |
|              | Front                 | $15255 \pm 623$                                                                   | $.343 \pm .015$                                                                   | $.253 \pm .001$                                                                   | $.540 \pm .004$                                                                     | $.015 \pm .000$                                                                     | $.690 \pm .041$                                                                     |
| $9 \times 9$ | Around                | $16744 \pm 117$                                                                   | $.372 \pm .002$                                                                   | $.257 \pm .001$                                                                   | $.534 \pm .001$                                                                     | $.015 \pm .000$                                                                     | $.728 \pm .009$                                                                     |
|              | Front                 | $15211 \pm 1416$                                                                  | $.342 \pm .033$                                                                   | $.253 \pm .001$                                                                   | $.541 \pm .007$                                                                     | $.015 \pm .000$                                                                     | $.664 \pm .050$                                                                     |

through the  $3 \times 3$  Front configuration outweighs that of seeing one pixel behind in the  $3 \times 3$  Around configuration, which gives most similar behavior to the Blind setting.

## F.7 VARYING ATTACK STRENGTH

This ablation tests whether the predation contribution depends on the default attack lethality in the  $256 \times 256$  **Ocean** world. We evaluate the following attack strengths: 1) Full Attack: deals 10 HP of damage, instantly killing any agent in the attack radius 2) Half Attack: deals 5 HP of damage 3) Quarter Attack: deals 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 were run with the Full Attack setting.

Reducing attack strength from Full Attack to Half Attack substantially reduces both attack frequency and attack effectiveness: attacks per agent and homicides per attack drop considerably (Figure 22). As a result, populations with lower attack strength grow larger, with agents moving and eating more frequently, resulting in greater biomass utilization. Further reducing strength to the Quarter Attack setting yields diminishing returns in terms of further reduction in violence and facilitation of additional population growth. Intuitively, when attacks do less damage, a homicide requires more attacks per agent, reducing the lethality of each attack. However when the lethality is reduced, attacking behavior becomes less useful and therefore less likely to be selected for in the agent population. Agents with lower attack strengths therefore do not become as skillful at attacking and spend more time foraging for food.



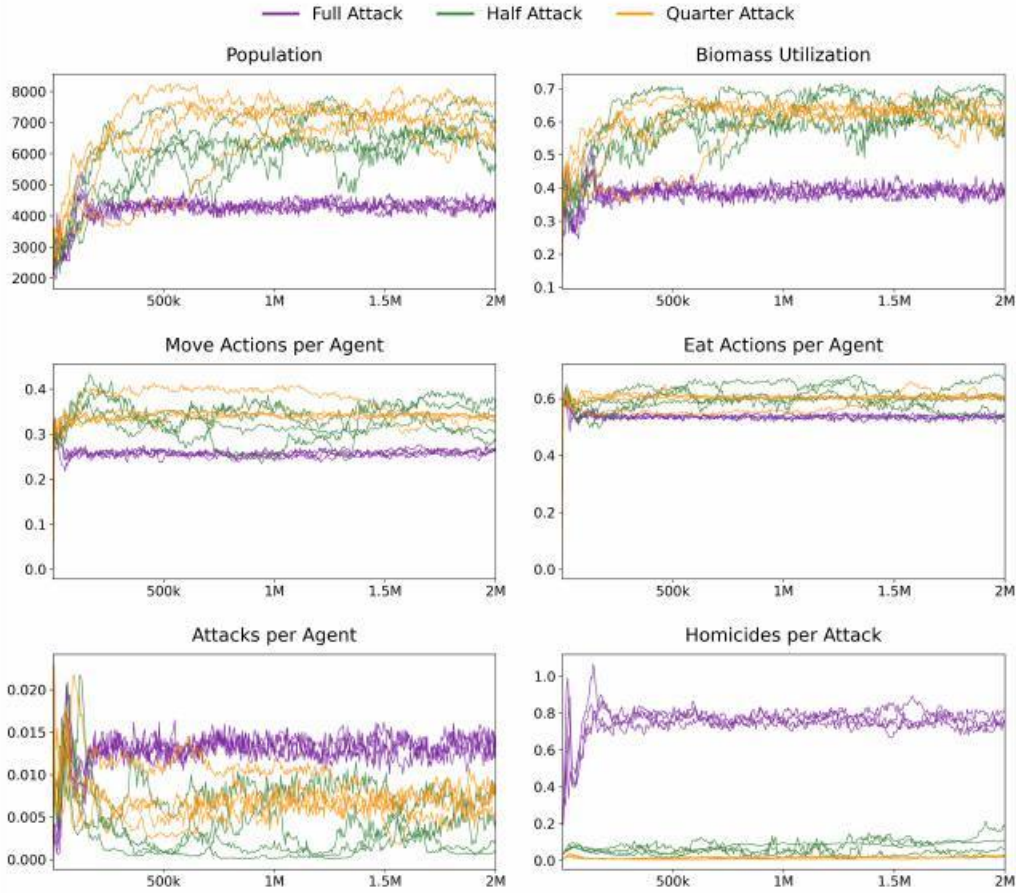


Figure 22: Attack-strength ablation for predation and foraging dynamics. Population and action-frequency metrics are shown over 2M steps in a  $256 \times 256$  **Ocean** world for (RCV+A) agents across Full Attack, Half Attack, and Quarter Attack settings, with four seeds per setting.

## F.8 VARYING AGENT HP

This ablation tests whether the main foraging and predation patterns depend on the default child HP and healing assumptions. The healing rate parameter determines how much HP an agent can regenerate per healing step, which converts available energy to HP. Agents can influence healing only by maintaining enough energy.

We study two settings: 1) Full HP: child HP starts at 5 and healing rate is 1 HP per 0.1 energy 2) Low HP: child HP starts at 1, with healing rate 0.5. Max agent HP is 10 in all settings. The main experiments (Section 4) were run with the Full HP setting.

Across the two settings, the main contribution metrics are largely stable: homicides per attack, eat action frequency, and biomass utilization show no significant difference (Figure 23). In the Full HP setting, populations grow slightly larger, while in the Low HP setting, agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must focus on foraging early on in their lifetime in order to grow to full size, which would explain this difference.

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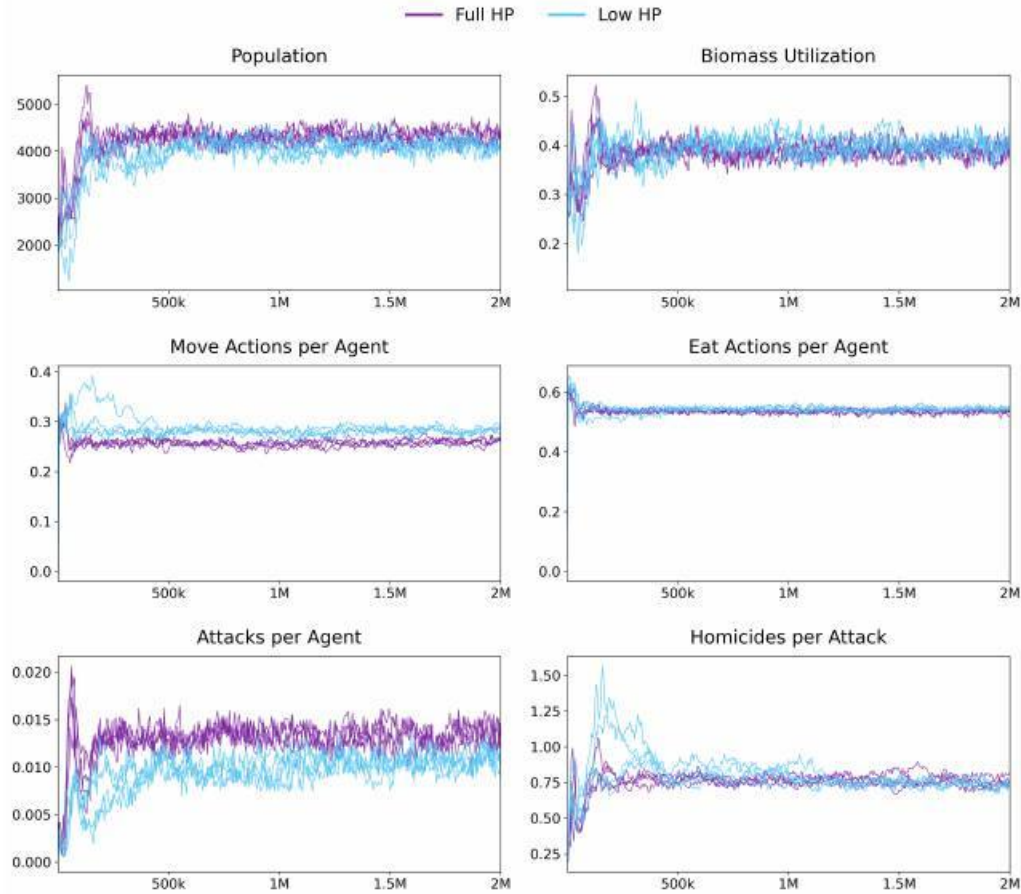


Figure 23: Agent-HP ablation for foraging and predation dynamics. Population and action-frequency metrics are shown over 2M steps in the 256x256 Ocean world for (RCV+A) agents across Full HP and Low HP settings, with four seeds per setting.